- *After page handler execution*—If you don't short-circuit the page handler execution, the page filter runs a third and final time after the page handler has executed. At this point you could customize the `IActionResult` returned by the page handler before the result is executed.

The triple execution of page filters makes it a bit harder to visualize the pipeline, but you can generally think of them as beefed-up action filters. Everything you can do with an action filter, you can do with a page filter, and you can hook in after page handler selection if necessary.

TIP Each execution of a filter executes a different method of the appropriate interface, so it's easy to know where you are in the pipeline and to execute a filter in only one of its possible locations if you wish.

One of the main questions I hear when people learn about filters in ASP.NET Core is "Why do we need them?" If the filter pipeline is like a mini middleware pipeline, why not use a middleware component directly, instead of introducing the filter concept? That's an excellent point, which I'll tackle in the next section.

21.3 Filters or middleware: Which should you choose?

The filter pipeline is similar to the middleware pipeline in many ways, but there are several subtle differences that you should consider when deciding which approach to use. The considerations are essentially the same as those for the minimal API endpoint filter I discussed in chapter 5. MVC filters and middleware are similar in three ways: